Equation of a tangent from a graph

1

a

i $(1, 0)$ ii 2 iii $y = 2x - 2$

b

i $(1, -3)$ ii -2 iii $y = -2x - 1$

c

i $(2, 2)$ ii -1 iii $y = -x + 4$

d

i $(4, 0)$ ii 1 iii $y = x - 4$

e

i $(2, 3)$ ii $\frac{1}{2}$ iii $y = \frac{x}{2} + 2$

2

a

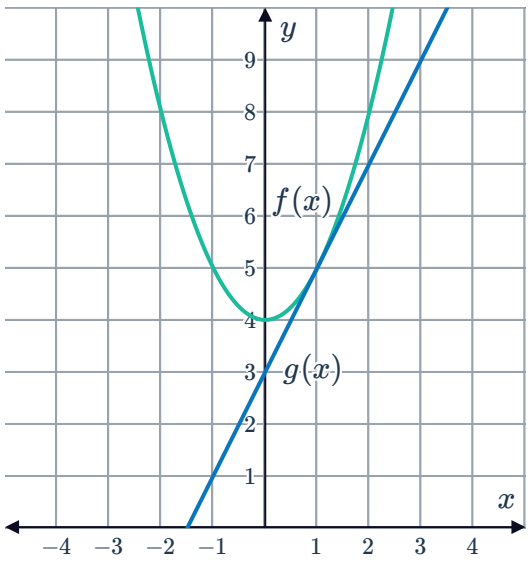
i $(-1, -6)$ ii 3 iii $y = 3x - 3$ iv $x = 1$

b

i $(1, -4)$ ii -3 iii $y = -3x - 1$ iv $x = -1$

3

a

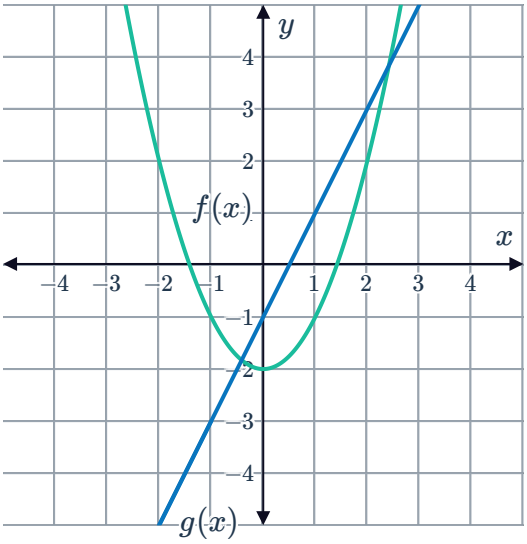


b Yes, because the line $g(x)$ touches the curve $f(x)$.

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4

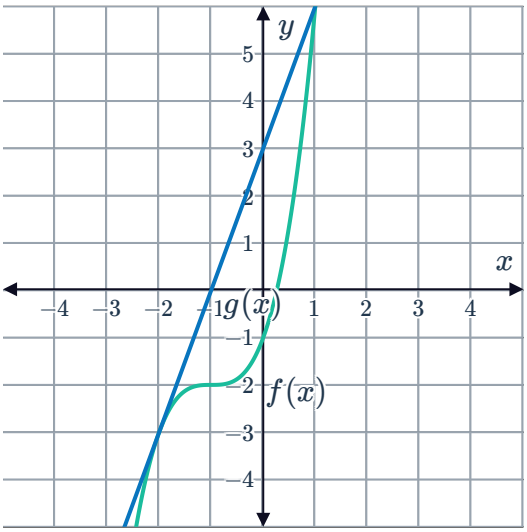
a



b No, because the line $g\,(x)$ crosses the curve $f\,(x)$ twice.

5

a

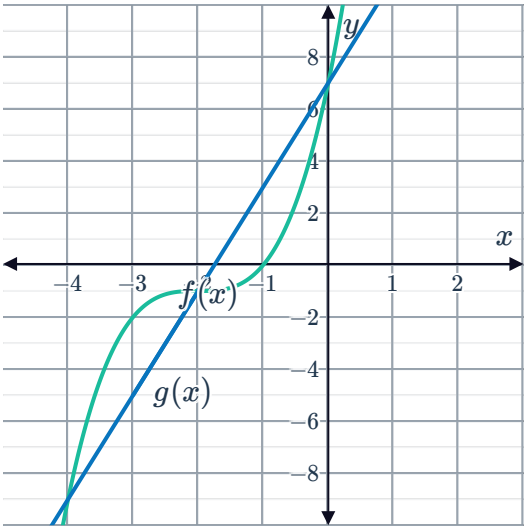


b Yes, because the line $g\,(x)$ touches the curve $f\,(x)$ at $x = -2$.

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6

a

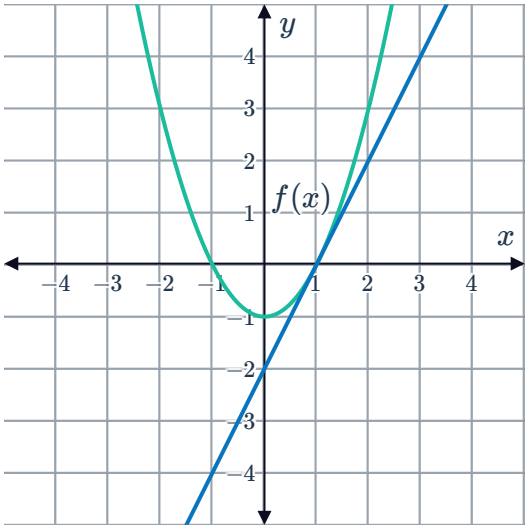


b No, because the line $g\,(x)$ crosses the curve $f\,(x)$ three times

7

a 2

b

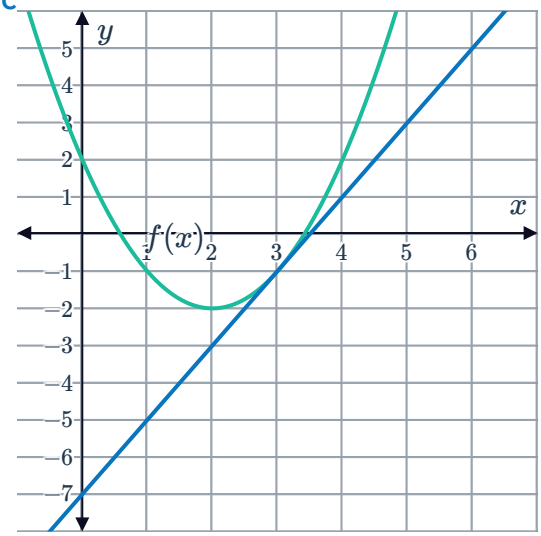


8

a 2

b $(2, -2)$

c



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Equation of a tangent at a given point

9 $x = 2$

10

a $f'(-1) = -2$ **b** $y = -2x - 1$

11

a $f'(-1) = 3$ **b** $y = 3x + 2$

12

a $f'(-3) = 6$ **b** $y = 6x + 9$

13

a $f'(-2) = -4$ **b** $y = -4x - 4$

14

a $f'(2) = 12$ **b** $y = 12$

c $y = 12x - 12$

15 $y = -10.1x + 8.4$

Applications

16

a

i $x = 2$ **ii** $y = 4$

b

i $x = 2$ **ii** $y = 8$

17 $x = \frac{3}{4}, x = -2$

18

a $\frac{dy}{dx} = 2x + b$ **b** -5 **c** $b = -23$ **d** $c = 79$

19

a $x = 4$ **b** $(4, 36)$

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20 **a** $x = 15, -11$ **b** $(15, 2025)\,,\,(-11, -2057)$

21
a $x = 2, -4$ **b** $(2, -16)\,,(-4, 62)$

22
a $x = 2, x = -6$ **b** $(2, 2)\,,(-6, 82)$

23 $k = 40$

24
a $\dfrac{dy}{dx} = 8x - 5$ **b** $x = \dfrac{5}{8}$

25 $\theta = 87^\circ$

26 $a = 3, b = 5$

27 $a = 2, b = 3, c = 0, d = -28$

28
a $a = 900$ **b** $b = 90$

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